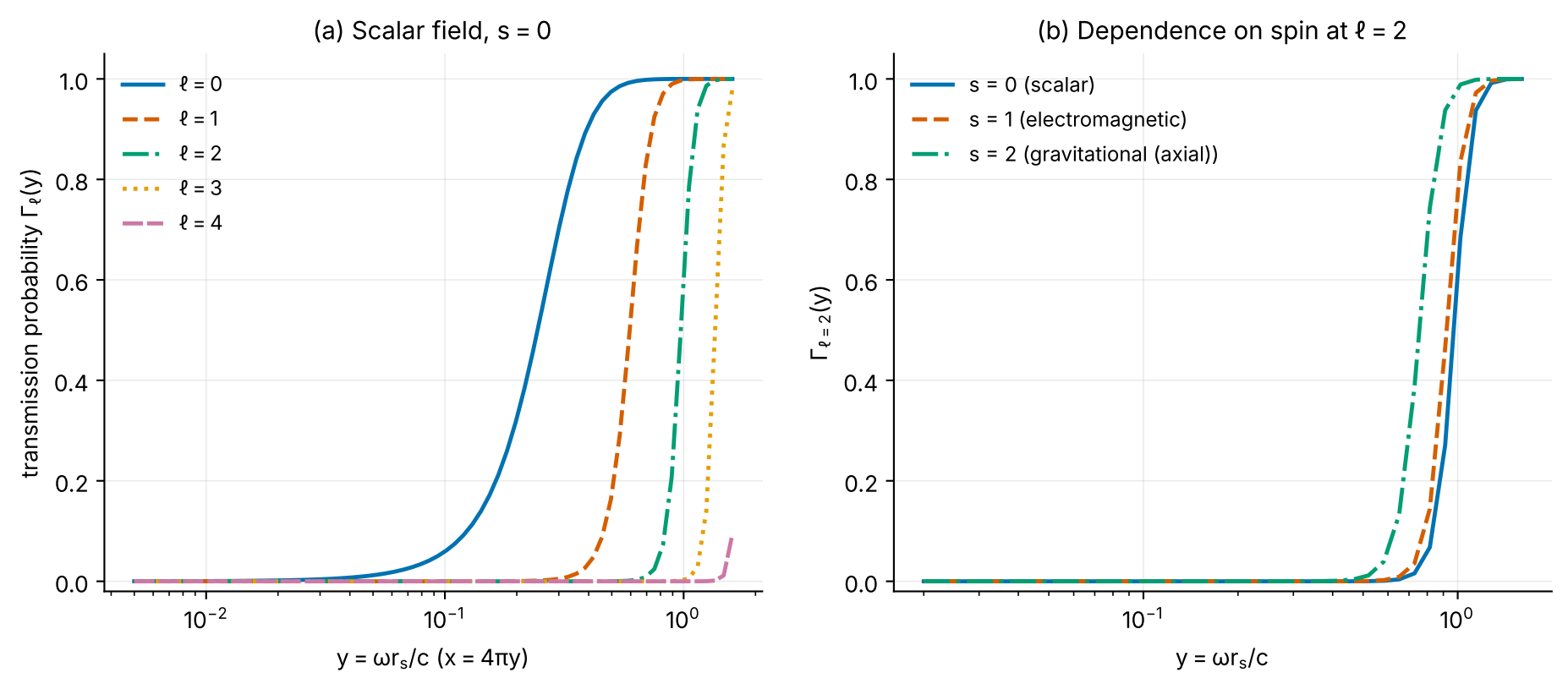
(a) Scalar field, s = 0
transmission probability $\Gamma_\ell(y)$
$\ell = 0$
$\ell = 1$
$\ell = 2$
$\ell = 3$
$\ell = 4$
1.0
0.8
0.6
0.4
0.2
0.0
10^{-2}
10^{-1}
10^{0}
$y = \omega r_s/c$ ($x = 4\pi y$)
(b) Dependence on spin at $\ell = 2$
$\Gamma_{\ell=2}(y)$
s = 0 (scalar)
s = 1 (electromagnetic)
s = 2 (gravitational (axial))
1.0
0.8
0.6
0.4
0.2
0.0
10^{-1}
10^{0}
$y = \omega r_s/c$